

Venkata Ram Sagar Konagandla

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EDUCATION

Texas A&M University at Qatar

Graduated May 2026

B.S. Petroleum Engineering — Undergraduate Research Scholar 2026

Thesis: *Experimental Rock Deformation and Digital Rock Physics Investigation of the Poro-Mechanical Evolution of Carbonate Rocks* — accepted; under embargo, public release forthcoming; journal manuscript in preparation

EXPERIENCE

Halliburton — Drilling & Evaluation

June 2025 — August 2025

MWD/LWD Operations Intern — Doha, Qatar

- Audited pre-deployment documentation packages for MWD/LWD tool crates, ensuring proof-of-readiness compliance and complete traceability before field dispatch.
- Authored tool failure reports by synthesizing operational and maintenance records to identify failure patterns and support root-cause analysis and corrective action workflows.
- Designed and implemented an automated tracking system for MWD/LWD tool readiness status, reducing filing errors and improving pre-deployment traceability across multiple crate inventories.

Texas A&M University at Qatar

Sept 2023 — May 2026

Undergraduate Research Assistant — Digital Rock Physics & Geomechanics (Dr. Seers & Dr. Fadlelmula)

- Designed and executed triaxial compression experiments on carbonate cores with stage-by-stage fluid-flow measurements and μ CT/DVC analysis to quantify internal strain field evolution and permeability changes under reservoir-representative stress conditions.
- Authored undergraduate thesis on poro-mechanical evolution of carbonate rocks (accepted; under embargo, public release forthcoming); currently preparing journal manuscript and conducting a comprehensive literature review to contextualize findings within the broader rock physics community.
- Ongoing collaboration planned to extend the experimental dataset to additional carbonate lithologies and stress regimes, working toward a generalized poro-mechanical framework for Middle East reservoir carbonates.
- Led ML dataset creation and model development for automated fault and fracture identification on outcrop imagery; findings published in *Scientific Data* (Nature Portfolio, 2024).
- Developed a second ML pipeline for lithology and sediment classification on thin section imagery, contributing to the group's automated petrographic characterization workflow.

Texas A&M University at Qatar

May 2021 — Nov 2022

Undergraduate Research Assistant — Fluid Dynamics (Dr. Azizur Rahman)

- Built and validated 3D geometry models using SolidWorks and ran CFD simulations in ANSYS to support fluid dynamics research on particle settling behavior; contributed to findings published in ASTFE (2023).

Amberloom Apparels

Dec 2024 — Oct 2025

Senior Marketing Intern

- Led influencer outreach across Qatar to grow brand visibility; contributed to strategic planning for product expansion and supported development of the brand's e-commerce platform.

PROJECTS

Algorithmic Trading Agent — Indian Equity Markets

2026 — Present

Co-developer — independent project, currently in testing

- Co-developing an automated trading agent for Indian equity markets, spanning market-data ingestion, signal generation, and trade-execution logic; currently validating strategy performance in a live testing phase before deployment.

SKILLS

Reservoir & Simulation: Petrel (reservoir modeling & property population), Eclipse E100 (black-oil & gas-condensate simulation), well log & seismic interpretation

Geomechanics & Rock Physics: Triaxial core testing, μ CT imaging, Digital Volume Correlation (DVC) analysis and meshing, poro-mechanical characterization of carbonates

Engineering Software: SolidWorks, ANSYS (CFD), Python (data processing & ML scripting)

Professional: Technical communication, root-cause analysis, research leadership, cross-functional teamwork

ACTIVITIES & LEADERSHIP

Society of Petroleum Engineers — Flow Assurance Club, TAMUQ

May 2022 — Dec 2024

President, Vice President, Secretary

- Organized lecture series, a sustainable engineering showcase, and multiple professional development events across three consecutive officer terms.

HBKU Student Enrichment Cycle

May 2024 — Oct 2024

Instructor — Innovative Design Optimization of Alternative Energy & Greenhouse

- Delivered ANSYS simulation workshops and wind energy optimization instruction to student cohorts at HBKU.